AI Lab Documentation

Security Alert Helper: A Generative AI Assistant for Junior SOC Analysts

Mazhar Mohammed Khursheed Ali | AIML-500 | Indiana Wesleyan University | Workshop 1, Assignment 1.4

Item	Details
Product	MKSecAlertHelper (custom chatbot)
Link	https://poe.com/MKSecAlertHelper
Platform	Poe (free plan), Prompt bot
Base model	Gemini 3 Flash
Target users	Junior Security Operations Center (SOC) analysts

1. Introduction

For this lab, I used generative AI tools and a design thinking process to build a simple AI assistant that solves a real problem. I chose a problem connected to my career goal of working in cybersecurity: helping new SOC analysts understand security alerts quickly and know what to do first. This document explains each design thinking stage, how I built and tested the chatbot, the challenges I faced, and what I learned.

2. Empathize: Understanding the Users

In 2025, I completed a SOC course at Full Stack Academy, where I worked with SIEM tools that collect and display security alerts. From that experience, I understood the challenges new analysts face:

- SIEM tools can produce a large number of alerts every day, and many are written in technical language.
- New analysts often are not sure how serious an alert is or which steps to take first.
- Senior analysts are busy, so juniors cannot always ask for help right away.
- Slow or wrong decisions on serious alerts can allow an attack to spread.

3. Define: Problem Statement

Problem statement: Junior SOC analysts need a quick and simple way to understand security alerts, judge how serious they are, and know the first steps to investigate, so they can respond faster and with more confidence.

4. Ideate: Exploring Solutions

I considered three possible AI assistant ideas:

Idea	Description	Decision
Security Alert Helper	Explains security alerts in plain language, rates severity, and suggests first investigation steps	Selected

Idea	Description	Decision
International Student Helper	Answers common questions about deadlines, registration, and campus resources	Not selected
Java Study Buddy	Explains object-oriented programming concepts with simple examples	Not selected

I selected the Security Alert Helper because it solves a real workplace problem, connects to my SOC training, and supports my goal of working in cybersecurity.

5. Prototype: Building the Assistant

5.1 Tools Used

- **Poe default Assistant:** used first to try out the instructions in a normal chat.
- **Poe Create Bot (Prompt bot):** used to build the permanent, shareable chatbot.
- **Gemini 3 Flash:** the generative AI model that powers the final bot.

5.2 Build Steps

1. Opened Poe and selected Create bot, choosing the Prompt bot type.
2. Chose Gemini 3 Flash as the base model.
3. Named the bot MKSecAlertHelper (Poe limits names to 20 characters).
4. Wrote clear and specific instructions describing how the bot should behave.
5. Added a greeting message that appears at the start of every conversation.
6. Published the bot and confirmed the link opened correctly.

5.3 Bot Instructions (Prompt)

The bot follows these instructions:

You are Security Alert Helper, an assistant for junior SOC analysts. When a user pastes a security alert or describes suspicious activity: (1) explain in simple language what the alert means; (2) rate how serious it likely is (Low, Medium, or High) and explain why in one or two sentences; (3) list 3 to 5 first steps to investigate it; (4) end with a short reminder to follow their organization's incident response policy. Rules: use short bullet points and keep answers under 200 words; if the alert is unclear or missing details, ask one question to get more information before answering; never tell the user to ignore an alert without investigating it; if the question is not about cybersecurity, politely say you only help with security alerts.

5.4 Greeting Message

Hi! I'm Security Alert Helper. Paste a security alert or describe suspicious activity, and I'll explain what it means, rate how serious it is, and give you first steps to investigate.

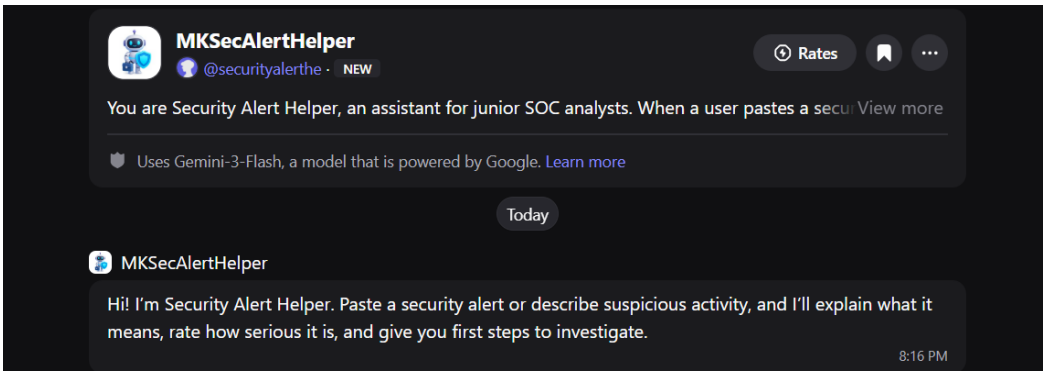


Figure 1. The published bot showing its instructions, Gemini 3 Flash model, and greeting message

6. Test: Results and Analysis

After the free daily points reset, I tested the bot with three realistic security alerts and one off-topic question to check whether it followed its instructions.

Test input	Bot response summary	Result
1. Multiple failed login attempts for admin account from an unknown IP	Identified a brute force attack and rated it High because admin accounts control critical systems. Gave four steps: check for successful logins, check IP reputation (e.g., VirusTotal), check whether other accounts were targeted (password spraying), and confirm with the administrator. Asked which service was targeted and ended with the policy reminder.	Passed. Followed all four instructions.
2. Antivirus detected a trojan on a user's laptop	Explained that a trojan is malware disguised as legitimate software that can steal data, spy on the user, or give an attacker remote control. Rated it High because trojans often act as a backdoor for ransomware or full network compromise. Gave five steps: check quarantine status, isolate the device, identify the source (downloads or emails), check network logs for unusual outbound traffic, and look up the file hash on VirusTotal.	Passed. Clear explanation, correct severity, five practical steps.
3. User logs in from two countries within one hour	Identified this as an "impossible travel" alert, meaning the account was used from two locations no person could travel between in that time. Rated it High because it suggests stolen credentials or a hijacked session. Gave steps: check for VPN use (a common false positive), verify MFA status, contact the user, and review account activity such as new email forwarding rules or large downloads.	Passed. Used the correct industry term and included false-positive checks.
4. Off-topic: What's a good pizza recipe?	Politely declined, explaining that it only helps with security alerts and cybersecurity investigations, and invited the user to share an alert instead.	Passed. Stayed within its defined scope.

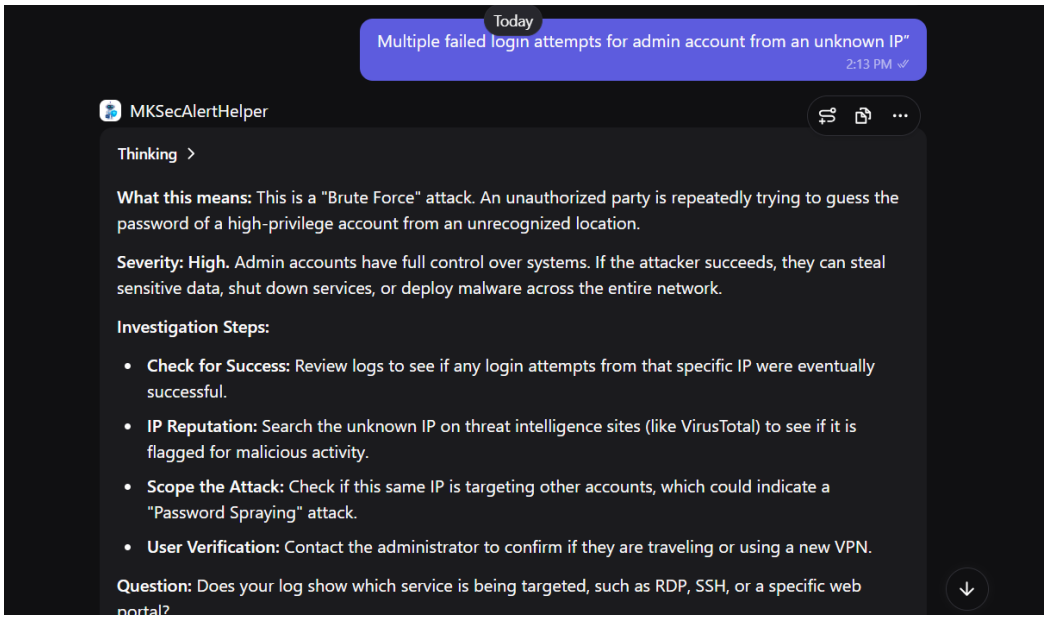


Figure 2. Test 1: brute force alert on an admin account

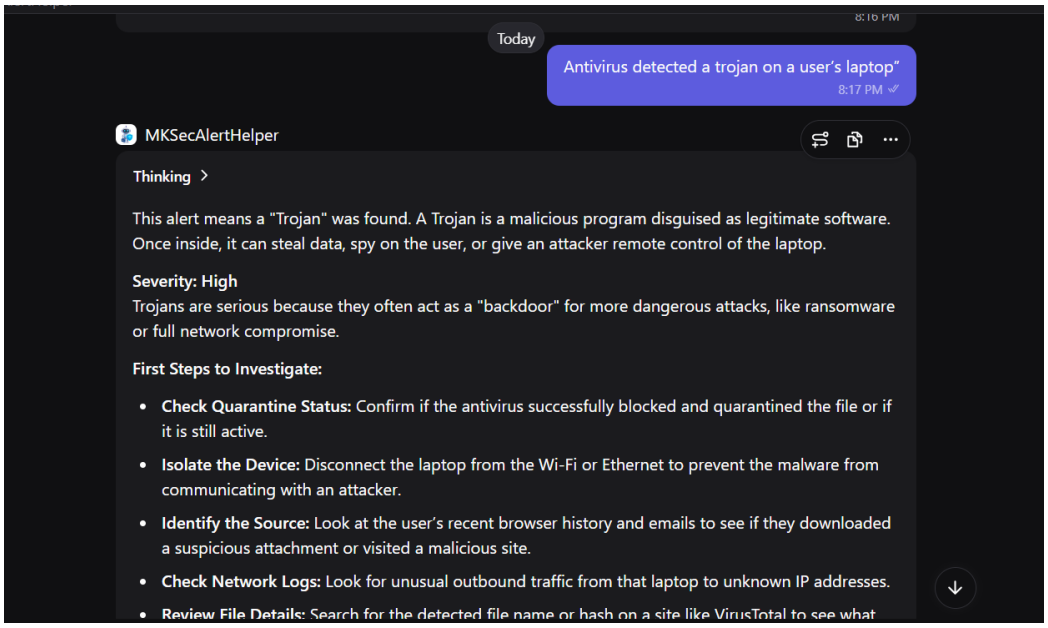


Figure 3. Test 2: trojan alert (answer continues beyond the screenshot)

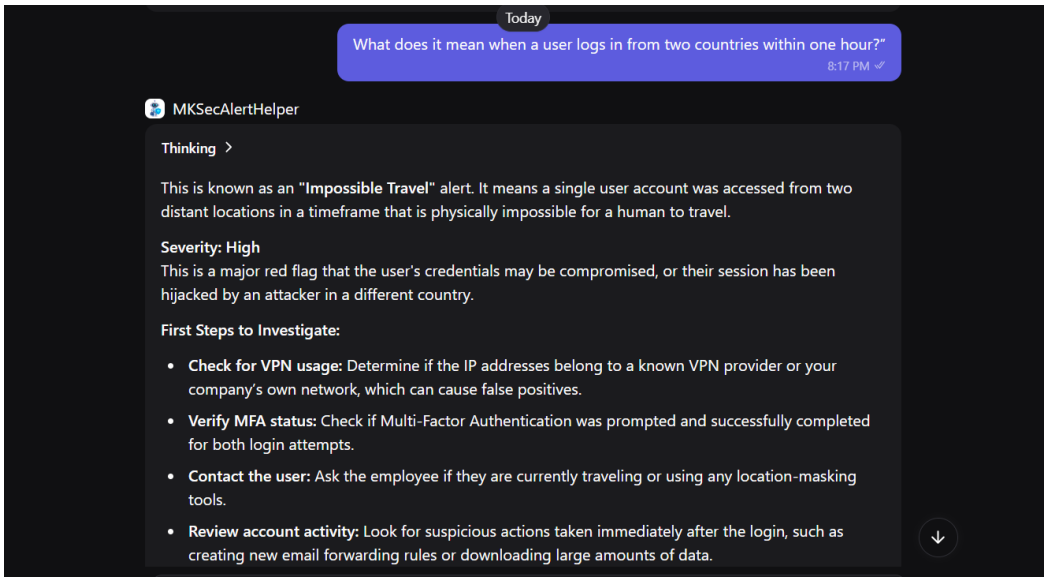


Figure 4. Test 3: impossible travel alert

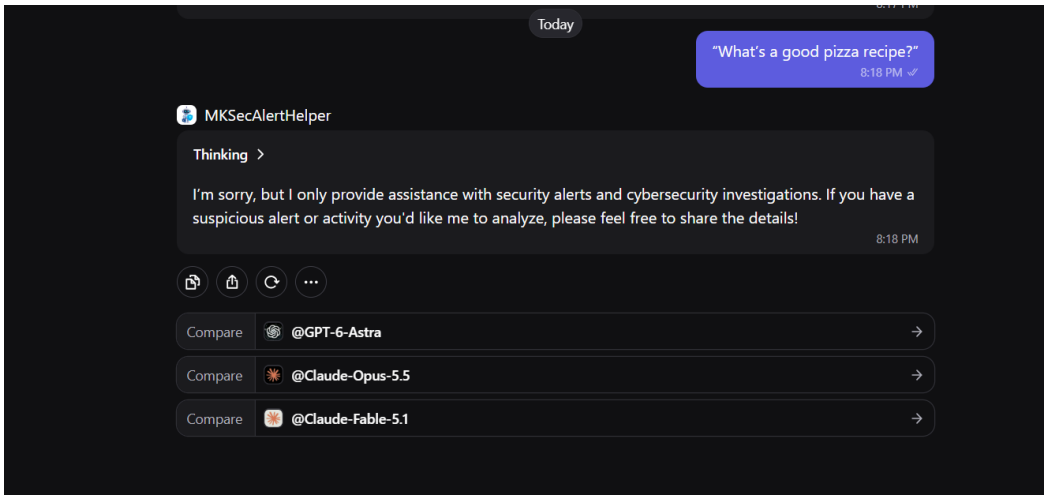


Figure 5. Test 4: off-topic question declined

6.1 Analysis of Results

In Test 1, the bot's answer was accurate, well-organized, and matched the required format. It used correct cybersecurity terms such as brute force and password spraying, while still explaining them simply. One observation is that the bot asked a follow-up question even though the alert had enough detail to answer. This is helpful but not always needed, so a future improvement would be to ask follow-up questions only when key details are missing.

In Test 2, the bot explained what a trojan is in simple terms and gave practical containment steps, such as isolating the device and checking the file hash on VirusTotal. In Test 3, it used the correct industry term, "impossible travel," and wisely included a check for VPN use, which is a common cause of false positives. In Test 4, it stayed within its defined scope and politely redirected the user, which shows the off-topic rule in my instructions worked.

Overall, the bot passed all four tests. It rated all three security alerts as High, which was reasonable for these scenarios, but a future test set should include a low-risk alert to confirm that the bot can also give

Low or Medium ratings. I also noticed that some answers, especially Test 2, were close to or above the 200-word limit I set, so the instructions could be tightened further. Finally, Gemini 3 Flash showed a "Thinking" step before answering, which slightly slowed responses but did not affect quality.

7. Challenges and Adaptations

Challenge	What happened	How I adapted
Bot was not created	I first pasted the instructions into a normal chat with Poe's default Assistant. It worked, but it did not create a bot or a shareable link.	I used Poe's Create Bot feature to build a permanent Prompt bot with its own link.
Name length limit	Poe allows a maximum of 20 characters for bot names.	I chose a shorter name, MKSecAlertHelper (15 characters).
Model too expensive	My first bot could not answer because its default model required more points than my free account had (233 points available).	I rebuilt the bot using Gemini 3 Flash, a faster, lower-cost text model. I avoided image models such as Nano Banana because the bot needs to write text.
Daily points ran out	After the first test, the free plan's daily points were used up and Poe asked me to upgrade.	Instead of paying, I planned the remaining tests after the daily reset to 300 points.

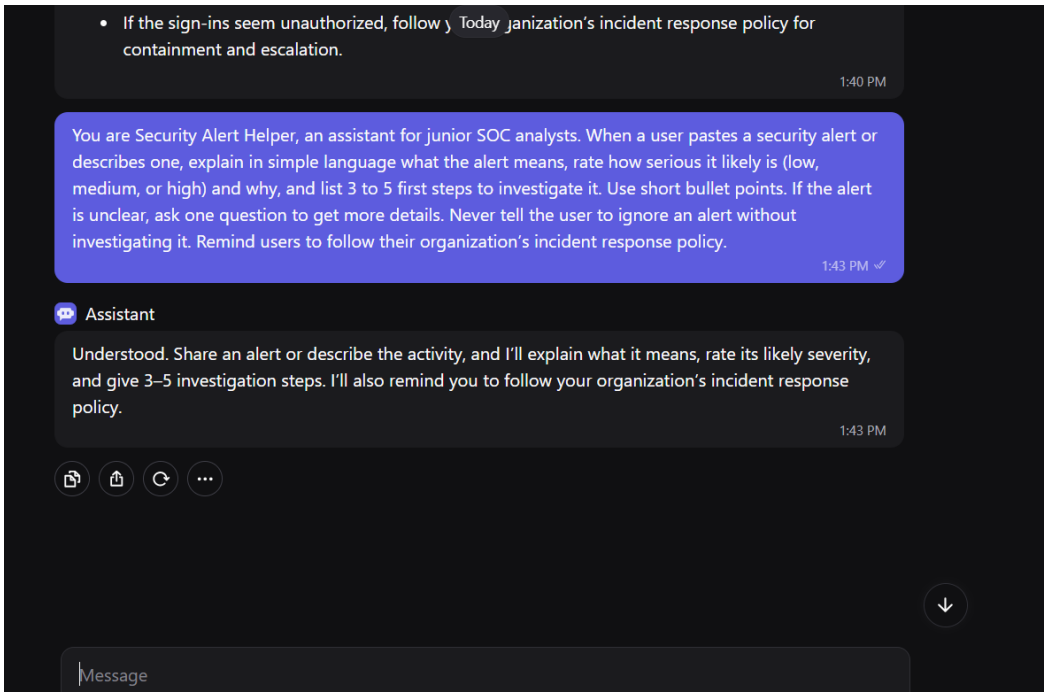


Figure 6. First attempt: instructions pasted into the default Assistant chat instead of a new bot

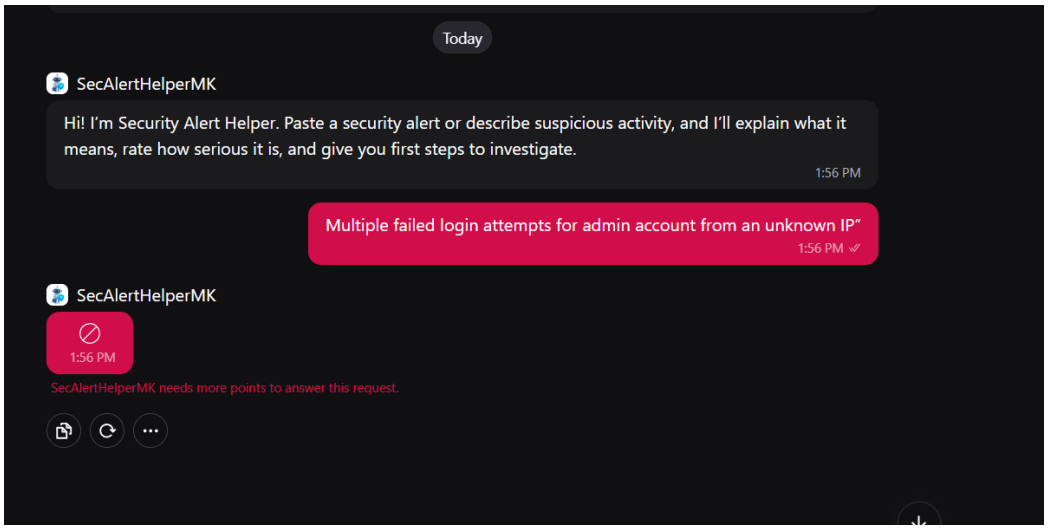


Figure 7. Error: the bot needed more points than the free account had

8. Insights and Lessons Learned

- Clear and specific instructions strongly control a chatbot's format and quality. The numbered steps in my prompt produced a consistent, well-organized answer.
- Choosing the right model matters. Larger models cost more, and lighter models like Gemini 3 Flash can be good enough for focused tasks.
- Free AI tools have limits, such as points and name rules, so planning ahead is part of building with AI.
- Design thinking kept the project focused on a real user need instead of only on the technology.
- AI assistants can support SOC analysts, but they should not replace human judgment or official incident response procedures.

9. Conclusion

Through this lab, I used generative AI tools to design, build, and test a working chatbot that helps junior SOC analysts understand security alerts. Following the design thinking process helped me understand the users, define a clear problem, and improve my solution when I ran into obstacles. This project strengthened my practical AI skills and connected them to my goal of working in cybersecurity.

Product link: https://poe.com/MKSecAlertHelper